

# Design and Development of an In-pipe Inspection Robot

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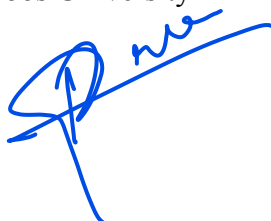
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## Abstract

Pipes are widely used engineering structures for transporting almost any fluid, from water to oil and gas. They have been one of the major components in many industries and are essential for life requirements around the world. The integrity of such structures is vulnerable to wear and tear, operational or manufacturing defect, natural disasters, aging, and physical damage from other sources, which can cause leaks, blockages, or damage to the piping.

As such, periodic or emergency inspections and maintenance are important operational requirements for those narrow and lengthy pipes. The complexity of this procedure stems from the pipes' diameters, lengths, the mediums inside, the complexity of the pipes' arrangement, etc. Using a tool that can save time and effort in a safe and efficient manner is essential. Therefore, robotic pipe inspection has been introduced.

In this study, a design of an inchworm-type in-pipe inspection robot is presented, and the resulting initial prototype is tested. Kinematic analysis was performed to analyze the robot's motion in vertical pipes, determining the optimized robot dimensions. Furthermore, the required minimum motor torque and track friction constant for movement in a vertical pipe were computed. The MATLAB optimization tool is used in optimizing robot parameters. By the end of the project, a proof of concept is reached, and a robot prototype is built and tested in a vertically oriented pipe.

The testing of the manufactured inchworm-type in-pipe inspection robot was conducted. The experimental robot performance can be enhanced by employing the optimized robot dimensions. The optimized robot dimensions are tabulated and discussed, and recommendations for improvement are stated.

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PREVIEW



# Chapter 1: Introduction

## Background

At present pipes or pipe-shaped components are widely used to transport fluids from water, oil and gas, to power plants fluids, and even beverages and many other flowing media. They have been some of the major components for many industries and life requirements around the world. During a pipe's life, this engineering structure is vulnerable to wear and tear, operation or manufacturing defect, natural disasters, aging, and physical damage from other sources that can cause leaks or piping damage.

Due to those threats of corrosion and erosion, leaks and operation stoppage can potentially affect the pipes, causing emergency plant shutdown. This triggers huge economic, infrastructure, and sometimes human loss. Inspection, cleaning, and monitoring techniques are critical components of this industry. In addition, the piping systems need to be checked periodically for their condition and integrity.

Pipe systems differ in their pipe diameters, lengths, the mediums inside, the complexity of the pipe arrangement, etc. This leads to pipes which are not always accessible by humans. Knowing the fact that robots are used in a wide variety of fields and tasks to reduce human intervention, especially in labor-intensive and hazardous work environments, inaccessible workplaces, or where the cost and time can be reduced intensively, the in-pipe inspection robots have been introduced to the piping and pipelines industry.

At present, many types of in-pipe inspection robots are available on the market. The main concern in designing a mobile platform or robot for traversing pipes is how to tackle varying pipe geometry such as pipe branching, pipe orientation, changing diameters, and pipe elbows. The robot has to be able to do its function in challenging environments because a wide variety of pipelines exists in the industry. Horizontal pipelines are relatively easy to inspect, but inclined or vertical pipes are more difficult to work with.

## **Problem Statement**

Current studies on in-pipe inspection robots are limited, despite the product having been available on the market a long time. Presently available products encounter significant challenge to travel in vertical pipes, such as bearing the weight of all their equipment and pulling their own cable. The aim of this study is to design a robot capable of ascending a vertical pipe while performing in-pipe inspection tasks.

With each type of robot, the important parameters to be considered differ. Optimizing these parameters will result in providing a robot that functions according to requirements and within budget constraints. In this study, an inspection robot is expected to travel in a vertical pipe against gravity. Analyzing the robot's movement will provide insight into the parameters that the designer needs to prioritize.

This study focuses on designing a prototype that will be tested in the lab. The inspection robot is expected to travel in a vertically oriented transparent pipe of diameter ranging from 10 to 14cm, as shown in Figure 1 below:

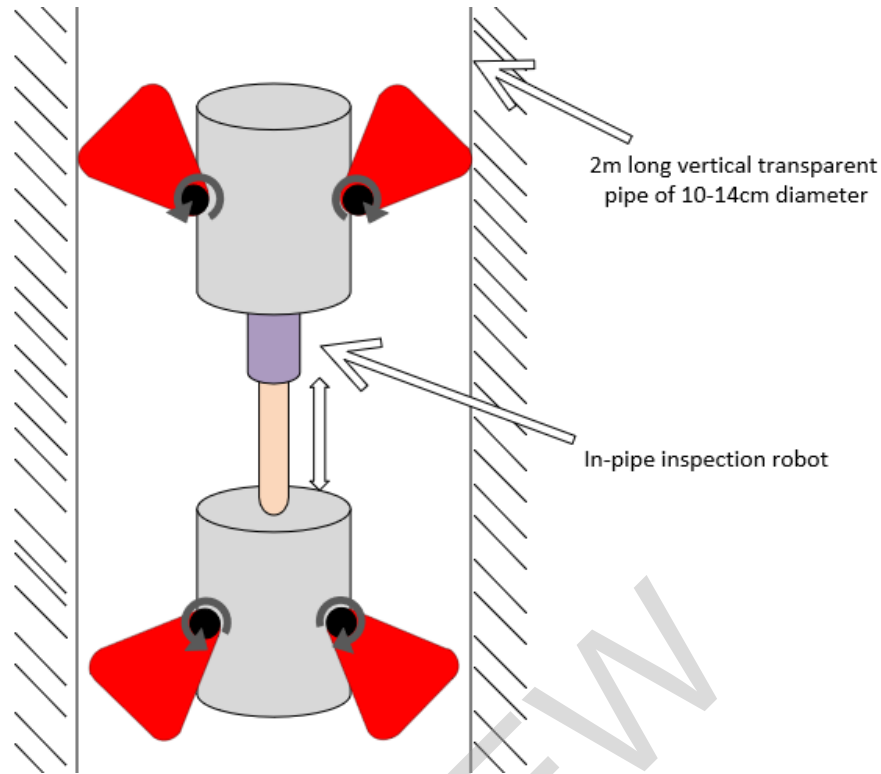


Figure 1: Experimental setup.

The robot should travel without slipping while accommodating the design payload of its equipment and pulling its cable. For simplicity, this research does not consider any fluid inside the pipe, as primary focus is on the mechanical performance of the robot.

## Study Objectives

The objectives were to design and optimize the preliminary design of an in-pipe inspection robot, considering the constraints outlined in the problem statement. The study includes literature review examining the limitations and challenges of various types of in-pipe inspection robots (IPIRs). Additionally, it involves selecting a suitable design aligned with the project objective, followed by its design, and fabrication, and testing of the manufactured robot. The tested robot should undergo refinement through optimization.

## **Chapter 2: Literature Review**

### **1. Mechanisms of In-pipe Inspection Robots**

The in-pipe robots can be categorized into three main categories: wheel-type, caterpillar-type and no-wheel type robots, and a number of subcategories as shown in Figure 2 and Figure 3 (Nayak & Pradhan, 2014). Other classifications of In-pipe inspection robots are based on their autonomy or controlling mechanism. Depending on the robots' control, they can be classified as non-autonomous robots, semi-autonomous robots, or fully-autonomous robots (Mamidi, 2015). The non-autonomous robot is a fully human-operator controlled type of robot mostly using a control cable. These robots are mostly used in plumbing operations. Secondly, although the semi-autonomous robot still needs operator intervention, it has some preprogrammed modules to achieve particular tasks once the operator gives it the order to start. Fully autonomous robots are still under development and more research is being done in this area (Mamidi, 2015). More explanation for different types of robots will be given in the next sections

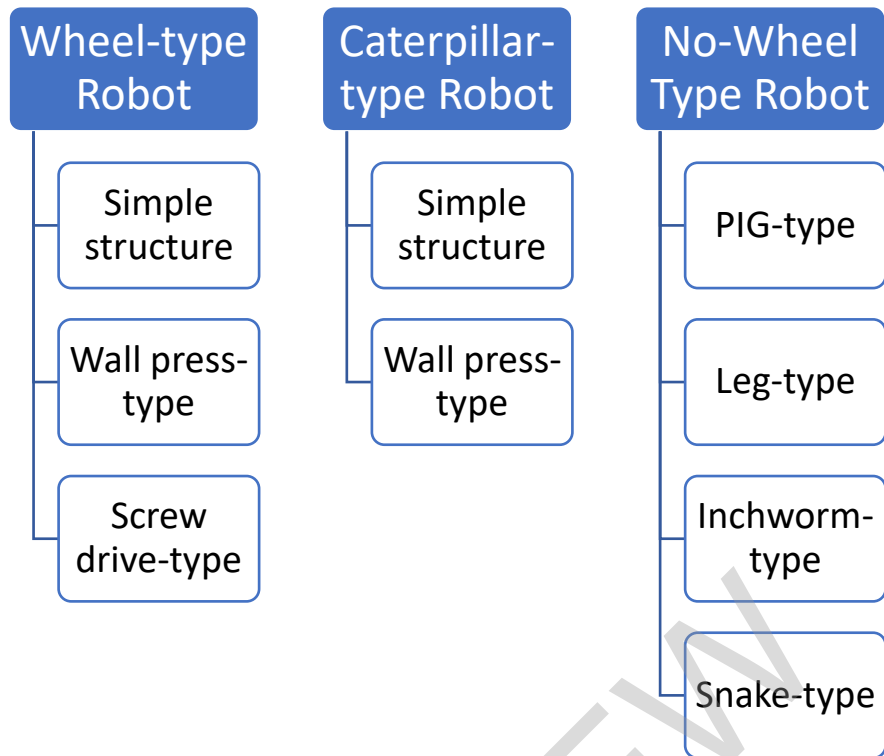


Figure 2: In-pipe inspection robot categories.

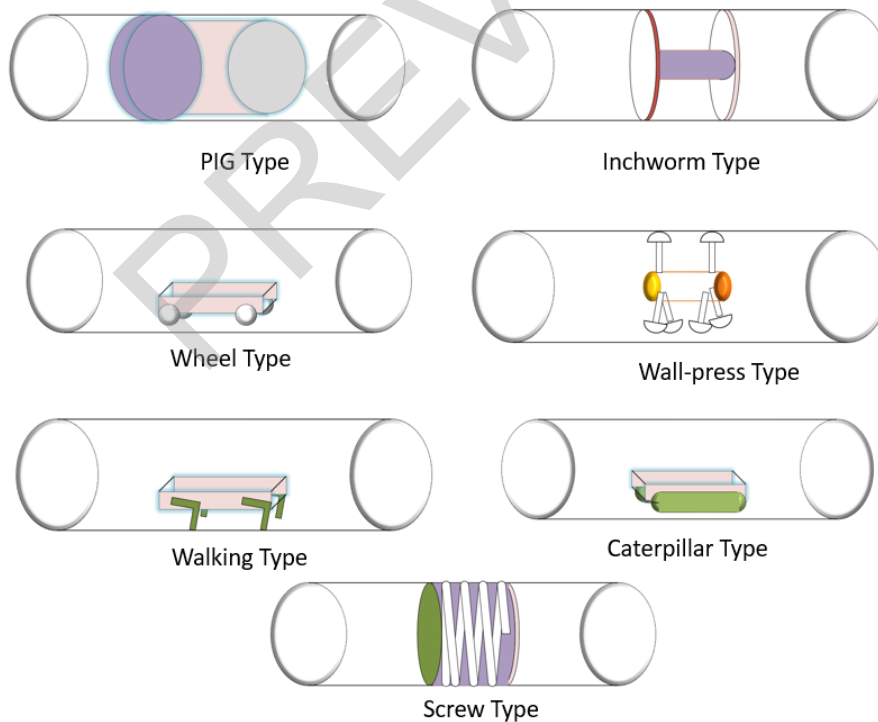


Figure 3: In-pipe inspection robot mechanisms.

### ***PIG-type Piping Robots***

The pipe inspection gauge (PIG) is a commercially popular robot. They do not generally use drive methods to maneuver along the pipeline and instead move inside the pipelines with the help of fluid movement and fluid kinetic energy. These types of robots can only be used effectively in high flow rate pipelines so that the flow can maneuver the robot. The main purpose is to carry out cleaning inside the pipelines. Moreover, added sensors and intelligent systems allow the robot to be used to assess long cross-country pipelines. These robots are equipped with an onboard power source only for sensors and the CPU. These robots can be equipped with instruments to perform a variety of tasks such as thickness measurement of pipelines, detection of corrosion, removal of slug and wax deposits, and opening choked pipelines. However, the PIG-type robot, as it is clear from Figure 4, almost has no controlled movement. It can rotate along the central axis, only with no forward and backward movement or speed controls (Marfatia, 2018); (Mishraa, D., Agrawalb K. K., Abbasa, A., Srivastavaa, R., Yadava, R.S., 2019); (Mamidi, 2015)



*Figure 4: PIG robot (Marfatia, 2018).*

### ***Wheeled and Caterpillar-type Piping Robots***

Figure 5 and Figure 6 show two of the simplest forms of the in-pipe inspection robot (IPIR): the wheeled and caterpillar-types of in-pipe inspection robots. They are used efficiently in the straight sections of pipes with relatively easy controls in terms of direction and speed. They can be used with difficulty in intersections and on bent paths and are more difficult when facing the T-section of the pipe, where the robots lose their stability. Simple forms of wheeled or caterpillar-type piping robots cannot travel in vertical sections of the pipeline (Marfatia, 2018). To enhance movement in bends and vertical pipes, more complicated wheel- and caterpillar-types have been developed, for example the integration of wall-press type of robots, as shown in Figure 6, or utilizing magnetic wheels as a means of vertical travel. The caterpillar-type robot offers a lot more grip than normal wheel-type robots. The wall-press type robots have the same advantages as wheeled and caterpillar-type robots in addition to the advanced capabilities to move in vertical pipe sections, and more easily maneuver through obstacles inside the pipes (Marfatia, 2018); (Zin, Sahari, Saad, Anuar, & Zulkarnain, 2012); (Mamidi, 2015).



*Figure 5: Wheel-type IPIR (Marfatia, 2018).*